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PR - 5.

Part A - conceptual Foundation.

Q1. What are Ensemble Learning methods and why are they effective?

⇒ * Ensemble Learning methods involve combining multiple base machine learning models (like decision tree) to create a single, more powerful predictive model.

* they are highly effective because aggregating multiple models helps balance out individual errors, which significantly improves overall predictive performance, stability and reduces the likelihood of overfitting.

Q2. Explain the diffrence between Bagging, Boosting and voteing?

⇒ Bagging :- ~~For~~ trains multiple models indipendently and in parallel on random subset of data. (e.g. random forest) averaging their predictions to reduce model variance.

* Boosting :- Trains models sequentially, where each New model specifically focuses on correcting the errors and misclassifications made by the previous ones.

* voteing :- combines predictions from multiple diffrent, pre-trained base Algorithms (e.g. KNN, SVM) and ~~et~~ determines the final output through a majority vote.

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Q3 What is Bias - variance trade-off, and how do ensemble methods address it?

=> The Bias - variance Trade - off is the balance a model must achieve between being too simplistc (High Bias / Underfitting) and being too complex and sensitive to noise (High variance (overfitting).

Ensemble methods address this by using specific techniques to tackle either extreme. Bagging primarily lowers variance by averaging out noice, while Boosting primarily Lower Bias by iteratively improving weak learners.

Q4. Explain voting classifiers and stacking Ensemble?

=> voting classifiers :- Aggregate the predictions of several independent base models and predicted the final class using either a strict majority rule (Hard Voting) or by averaging the predicted class probabiliteies (soft voting).

Stacking Ensemble :- Trains multiple base models and than uses their outputs as inputs features to train a New higher Level "meta model" that Learns the optimal way to combine those base predictions for the final output.

Q.5. compare AdaBoost, Gradint Boosting, Light GBM, and XGBoost conceptually?

=> * AdaBoost :- sequentially assigns higher weight to missclassified data points so subsequent models are forced to focus on the hardest - to - predict cases.

* Gridient Boosting :- Builds sequential models by using gradient decent to minimize a loss function, specifically fitting new treas to the residual errors of the previous ones.

* XGBoost :- A highly optimized and scalable implimantaction of grident Boosting that includes built-in L1/L2 regularization to actively prevent overfitting.

* Light GBM :- A fast, hight performance ~~grid~~ gradiant Boosting Framework optimized for Large dataset that uniqualy uses a leaf - wise (rather than ~~ea~~ Level wise) tree groth strategy.